GATE DA · DBMS practice book

80 original problems: P01–P40 in chapter order, then four mixed tests of ten 2-mark questions. Lettered parts belong to the same problem. Write intermediate rows, closures, node states or calculations. No answers appear in this book. Use the separate solutions guide after attempting each problem.

Chapter 1 · Relations, keys and constraints

P01 · Short answer

Student(ID, Name, Email) has declared candidate keys {ID} and {Email}. The current instance has 4 rows and distinct names. (a) List all candidate keys and all superkeys. Give its degree and cardinality. (b) Is {Name} a schema key? If a fifth row is added, which count changes?

P02 · MSQ plus variation

Parent has primary keys {1,2}. Child has (ID,ParentID) rows (10,1),(11,2),(12,NULL), with a nullable foreign key and no cascade action. Which operations are guaranteed safe for this foreign key alone? A: add a fresh parent 3. B: delete child 10. C: add child (13,9). D: delete parent 2. (b) Repeat D when ON DELETE SET NULL is specified and the child column remains nullable.

P03 · NAT plus variation

Customer has unique IDs {1,2,3}. Purchase has references {1,1,2,NULL}; its non-null references form a valid foreign key. (a) Count the inner join rows and Purchase LEFT JOIN Customer rows. (b) Add a purchase with reference 3. Recompute both counts. Does the inner join list each customer once?

Chapter 2 · Relational algebra and tuple calculus

P04 · NAT plus variation

X(P,Q)={(1,a),(2,a),(3,b)}, Y(Q,R)={(a,9),(a,10),(b,10)}. (a) List X natural-join Y, then select R=10 and project Q. How many tuples remain? (b) Remove (b,10) from Y and repeat. What is the product size |X×Y| in each case?

P05 · NAT plus variation

Distinct R tuples have join values {1,1,2,3}. Distinct S tuples have join values {1,1,1,2}. (a) Count equality-join pairs. (b) Keep just one S tuple per join value. Recount. Explain why an upper bound of |R| failed in (a).

P06 · Short answer

Enrolment={(1,M),(1,P),(2,M)}, Student={1,2,3}. (a) Divide Enrolment by Required={M,P}. (b) Make Required empty. Compare classical division, whose candidates come from Enrolment, with a no-missing-course query whose candidates come from Student.

P07 · Counterexample

(a) Give a one-name instance where Team minus (Defender union Forward) is empty but Team minus (Defender intersection Forward) is nonempty. (b) Give an instance where (Defender union Forward) minus Team is empty but the required Team coverage fails. Explain what each reversed containment tests.

P08 · NAT plus variation

X={(1,a),(2,b),(3,c)} with attributes (P,Q); Y={(1,10),(1,20),(2,30),(3,1)} with attributes (P,S). Output each x in X for which some y in Y has the same P and S>1. (a) Count output tuples and successful (x,y) witnesses. (b) Change >1 to ≥1. Recompute. Does an additional ∃z∈X with z.P=x.P change the output?

Chapter 3 · SQL: rows, groups and nested questions

P09 · NAT plus variation

Customer rows are (1,Asha),(2,Ravi); Purchase rows are (1,10),(1,20),(2,30). (a) Count SELECT Name over the matching join and SELECT DISTINCT Name over it. Also count customers selected by a correlated EXISTS on purchases. (b) Add Purchase (1,40). Recompute all three counts.

P10 · Short answer

Value contains 10,NULL,10,20. (a) Give COUNT(*), COUNT(Value), COUNT(DISTINCT Value), SUM(Value), AVG(Value). (b) Repeat on WHERE Value>100. How many result rows does the ungrouped aggregate have? How many groups does GROUP BY Value have on that empty input?

P11 · Query tracing

Sales(team,amount)={(A,10),(A,20),(B,7),(B,NULL)}. Evaluate SELECT team,SUM(amount) FROM Sales WHERE amount>8 GROUP BY team HAVING SUM(amount)>25. (a) Show rows after WHERE and the final groups. (b) Add (B,30) and repeat. Would COUNT(*) and COUNT(amount) agree before applying WHERE?

P12 · NAT plus variation

Balances are 5,5,10,15,18. Retain an account when the number of balances strictly above it is at least the number strictly below it. (a) List both counts per balance and give the output count. (b) Replace 15 by 10 and repeat. Treat tied accounts as separate rows.

P13 · Query comparison

Applicant IDs are 1,2,3. Rejected IDs are 2,NULL. (a) Compare WHERE ID NOT IN (SELECT ID FROM Rejected) with a matching-ID NOT EXISTS query. (b) Remove the NULL row and compare again. Also give the truth values of 5>ANY(empty) and 5>ALL(empty).

P14 · Query tracing

C={(1,Asha),(2,Ravi)}, P={(1,10),(1,20)}. (a) List C LEFT JOIN P ON C.ID=P.ID AND P.amount>15. Compare moving P.amount>15 to WHERE. (b) A self join has values 1,1,2 on each side. Count ordered pairs for A.value<B.value and for A.value≤B.value.

Chapter 4 · Dependencies, keys and normal forms

P15 · Closure

R(UVWXYZ) has U→V,U→W,V→X,WX→Y,WX→Z. (a) Compute (VW)+ and U+. Does VW→U follow? (b) Remove V→X. Recompute (VW)+ and explain which step stops working.

P16 · All keys

R(ABCD) has A→B,B→A,C→D. (a) Find all candidate keys and prime attributes. (b) Remove B→A and repeat. Show minimality, not only full closure.

P17 · NAT plus variation

Use R(ABCD) with A→B,B→A,C→D. (a) Count its superkeys. (b) Add attribute E and FD A→E; count again. What if E were added without any rule determining it?

P18 · Cover and equivalence

F={A→BC,B→C,AB→D}. (a) Find a singleton-right-side minimal cover. Check whether G={A→B,B→C,A→D} is equivalent to F. (b) Remove A→D from G. Give a closure witnessing the loss of equivalence.

P19 · Normal form

Assume atomic values and no nontrivial empty-determinant FDs. (a) R(ABC) has AB→C,C→B. Find all keys and its highest normal form. (b) Replace the FDs with A→B,B→C. Repeat. Give a specific violation of the next higher normal form.

P20 · Losslessness

R(ABC) has only A→B. (a) Test AB/AC and AB/BC for losslessness. Give a two-row counterexample for the failing split. (b) Add B→C. Does AB/BC become lossless?

P21 · Chase

R(ABCD), F={A→B,B→C}, is split into AB,BC,AD. (a) Build a chase tableau and decide losslessness. (b) Remove B→C and repeat. State the row or missing distinguished symbol that decides each result.

P22 · Preservation

R(ABC) has A→B,B→C. (a) Does AB/AC preserve all dependencies? List its implied local FDs. Is it lossless? (b) Repeat for AB/BC. Use B's closure under projected rules to explain the difference.

P23 · Construct and check

R(ABC) has AB→C,C→B. (a) Perform the BCNF split using C→B. Check each component's normal form, losslessness and preservation. (b) Perform 3NF synthesis from these FDs, remove contained relations and check whether a key relation must be added. Why can keeping ABC be acceptable?

Chapter 5 · ER models: turn a story into constraints

P24 · Bounds and feasibility

A has 3 entities and B has 2. Both participate totally in a binary relationship. (a) Give minimum and maximum relationship pairs for many-to-many, with examples attaining both. (b) Make the relationship many-to-one from A to B. Give the pair count. Is both-side total participation feasible if A has only 1 entity?

P25 · Schema mapping

Students each belong to one department, can enrol in many courses, have several phone numbers and can own weak Dependents identified by DependentName within a student. Courses can have many students. (a) Give the tables, primary keys and references for Student, Department, Course, Phone, Enrolment and Dependent. (b) If each student instead has at most one phone, which table can disappear?

P26 · NAT plus variation

E1 and E2 are independent strong entity types with single-attribute keys. E2 has one multivalued attribute. Their relationship is many-to-many and both sides are total; no extra dependencies allow merging. (a) Give the minimum number of 3NF tables and their purposes. (b) Make the relationship many-to-one from E1 to E2 with E1 total. Recount and state the new reference location.

Chapter 6 · Files, indexes and B+ trees

P27 · NAT plus variation

Store 101 fixed 100-byte records unspanned in 1024-byte blocks. (a) Give records per block and required blocks. Why is ceil(10100/1024) wrong here? (b) Reserve a 50-byte header in every block and repeat.

P28 · Multilevel index

10,000 unspanned 100-byte records use 1000-byte blocks. Every index entry is 20 bytes at every level. (a) A sparse lowest level has one anchor per data block. Give block counts at each index level and uncached lookup I/Os including one data block. (b) Replace it with one dense entry per record and repeat.

P29 · Capacity

A B+ node has 1024 usable bytes; separator keys are 8 bytes and child pointers 4 bytes. Leaf entries have an 8-byte key plus a 12-byte record pointer, with a 4-byte leaf-link overhead. (a) Find maximum internal fanout and leaf capacity. (b) Double child-pointer bytes to 8; leave record pointers and leaf link unchanged. Recompute both capacities.

P30 · Height and occupancy

Maximum leaf capacity is 50 entries and maximum internal fanout is 100. (a) Find minimum height for 10,000 records and for 5000, both in levels and edges. (b) A 3-level tree has minimum nonroot fanout 50, minimum leaf occupancy 25 and minimum root fanout 2. Give its minimum and maximum record capacities.

P31 · Insertion and deletion

Use minimum-right-subtree separators. Leaves hold at most 2 keys and internal nodes at most 3 children. Split an overflowing 3-key leaf into 2 keys left and 1 right; split an overflowing 3-separator internal node by promoting its middle key. Start root [7], leaves [1,4] | [7,10]. (a) Insert 2, then 3; draw every tree state. (b) Delete 10, then 7. Nonroot leaves need at least 1 key and internal nodes at least 2 children. Borrow if a sibling can lend; otherwise merge and collapse a redundant root.

P32 · Choice and cost

(a) For each predicate, say whether standard hashing, an ordered B+ index, or both support a direct lookup: Rating>3.4; Genre='Comedy'; exact MovieID. (b) A B+ path has 3 levels and its root is cached. Count I/Os for one leaf lookup plus one data-page fetch, then for a covering lookup. A range touches 5 leaf pages and 6 data pages; count total I/Os without rereading the first leaf.

Chapter 7 · Warehouses, hierarchies and measures

P33 · Grain and multiplicity

Sale(LineID,ProductID,CityID,DayID,Amount) has one row per sale line and three flat dimension tables. (a) State the grain and count tables in its star. Move Product's category attributes to a separate Category table: recount. (b) Product A has sale amount 100 and tags red,new; product B has amount 50 and tag red. What is the sale sum before and after joining every tag? Explain the error.

P34 · Classification

Classify: Country→State grouping; City→State grouping; fixing Country='IN'; restricting several Product and Country values; exchanging the report's row and column axes. (b) Does a larger output row count, without knowing the grouping/filter changes, prove drill-down? Give a counterexample.

P35 · CUBE and hierarchies

Facts(Product,Country,Sales)={(A,IN,10),(B,IN,20),(A,US,30)}, with non-null dimensions. (a) Give the number of CUBE grouping sets and total output rows. Add (B,US,40) and recount. (b) Geography has 4 levels including ALL, Product 3, Time 4. How many hierarchy cuboids are possible under independent linear hierarchies?

P36 · Roll-up measures

(a) A group of 2 observations averages 100 and a group of 8 averages 50. Compute the combined mean and name the summaries needed. (b) Customer sets in two groups are {1,2} and {2,3}; find the overall distinct count. Explain why subgroup counts alone are insufficient. Can account balances be summed across accounts and across dates? Classify AVG's computation and balance's additivity.

Chapter 8 · Data types and preprocessing

P37 · Classification and calculation

(a) Classify department code, Low/Medium/High rating, Celsius temperature and mass in kg as nominal/ordinal/interval/ratio. (b) For x=60,min=20,max=80,mean=50,SD=10, compute min-max to [0,1], min-max to [-1,1] and z-score. For decimal scaling use max absolute value 80 and the smallest j with max|x|/10^j<1. (c) What breaks if all observations equal 7?

P38 · Bins and smoothing

Data are 0,1,2,3,8,9. (a) Use intervals [0,3),[3,6),[6,9] and replace each observation by its bin mean. Give memberships and the smoothed sequence. (b) Use equal-frequency bins of two sorted observations each and repeat. Explain how assigning bin labels instead would change the output type.

P39 · Sampling

Strata A and B have population sizes 90 and 10. Five sampled observations from each stratum have means 10 and 30. (a) Estimate the population mean with appropriate weights and compare the unweighted sample mean. (b) Count two ordered draws from five IDs with and without replacement. If a design selects villages and includes all their households, what is the first-stage sampling unit?

P40 · Compression

(a) Original size is 1000 bytes. Encoded payload is 600 bytes and header 100 bytes. Give total encoded size, original/encoded ratio and percentage saved. (b) Encode AAAABBCD as (symbol,count) runs with 8 bits per symbol, 8 per count and an 8-bit header. Compare encoded and original sizes. Why can no lossless code shorten every two-bit input?

Mixed tests · Database Management and Warehousing

Four original tests, each with ten 2-mark questions. Total: 20 marks per test. This is a preparation target, not a promised GATE section allocation.

Try each test in 40 minutes without notes, then spend 20 minutes correcting it. Score 2 for a completely correct answer and 0 otherwise. MSQ requires all and only the correct choices. This practice scoring scheme has no negative marks. Keep tests unopened until the relevant chapters are finished. Answers and recipe labels appear only in the separate guide.

Test 1 · 20 marks

T1Q01 · MCQ · 2 marks

Payment has a primary key PaymentID and a valid non-null BuyerID foreign key to Buyer's primary key. Which must hold? A: BuyerID is a Payment candidate key. B: the inner join of Payment to Buyer on BuyerID has exactly as many rows as Payment. C: deleting any Buyer is safe. D: distinct current buyer names make Name a schema key.

T1Q02 · NAT · 2 marks

R(A,B)={(1,2),(1,3),(2,3)}; S(B,C)={(2,9),(3,9)}. Count tuples in projection A of selection C=9 on the natural join.

T1Q03 · NAT · 2 marks

T(v) contains 10,10,NULL,30. What is COUNT(*)−COUNT(v) over all rows?

T1Q04 · MCQ · 2 marks

R(ABCD) has A→BC,C→A,D→B. Which is its complete candidate-key set? A: {A,C}. B: {AD}. C: {AD,CD}. D: {AD,BD,CD}.

T1Q05 · MSQ · 2 marks

R(ABCD) has A→B,B→CD and is decomposed into AB,ACD. Select all true statements. A: the split is lossless. B: it preserves every original FD. C: both components are BCNF. D: the original is BCNF. Assume atomic values.

T1Q06 · NAT · 2 marks

Four projects each have exactly one engineer as leader. Three engineers each lead at least one project. How many Project-Leader relationship pairs are stored?

T1Q07 · NAT · 2 marks

An internal B+ node has 512 usable bytes. Keys and child pointers each use 8 bytes. What is its maximum number of child pointers?

T1Q08 · NAT · 2 marks

Non-null facts with dimension pairs are (A,1),(B,1),(B,2),(C,2). How many rows does CUBE over both dimensions produce? Each pair has one measured fact.

T1Q09 · NAT · 2 marks

A value is 15, the population mean is 9 and the given standard deviation is 3. Compute its z-score.

T1Q10 · MSQ · 2 marks

Select all true statements. A: sum and count suffice to combine subgroup averages. B: arbitrary subgroup distinct counts can always be added. C: daily account balances are additive over dates. D: three ordinary CUBE dimensions create eight grouping sets.

Test 2 · 20 marks

T2Q01 · MSQ · 2 marks

Select all true statements. A: a foreign key is necessarily unique. B: a SQL primary key is non-null. C: two candidate keys can have different sizes. D: distinct values in a displayed instance prove a schema key.

T2Q02 · NAT · 2 marks

Enrolment={(1,a),(1,b),(2,a),(3,b)} and Required={a,b}. How many candidate IDs from Enrolment satisfy every required course?

T2Q03 · NAT · 2 marks

A(id) has 1,2,3. R(id) has 2,NULL. Count rows returned by SELECT id FROM A WHERE id NOT IN (SELECT id FROM R).

T2Q04 · NAT · 2 marks

Balances are 2,4,4,9. Keep a row only when the number strictly above it is strictly greater than the number strictly below it. How many rows survive?

T2Q05 · MCQ · 2 marks

Atomic R(ABCD) has AB→C,C→D and no other independent FDs. What is its highest normal form? A: 1NF. B: 2NF. C: 3NF. D: BCNF. Exclude nontrivial empty determinants.

T2Q06 · NAT · 2 marks

Owner has a multivalued Skill. Dependent is weak under Owner, identified by Name within its owner. Hobby is a strong entity. Dependent and Hobby have a many-to-many relationship. Under this standard atomic 3NF mapping with no extra merging rules, how many tables are needed?

T2Q07 · NAT · 2 marks

950 unspanned 90-byte records use 1000-byte blocks. A sparse index has one 20-byte entry per data block, also in 1000-byte blocks, and higher levels index blocks below. Count uncached lookup I/Os including one data fetch.

T2Q08 · MCQ · 2 marks

A report grouped by Month changes to Year totals. Which operation is this? A: drill-down. B: roll-up. C: dice. D: pivot.

T2Q09 · NAT · 2 marks

Sorted observations 0,2,4,6 are placed in two equal-frequency bins and smoothed by bin means. What value replaces the original 4?

T2Q10 · NAT · 2 marks

AAAAABBBBB originally uses 8 bits per character. A run uses 8 bits for its symbol and 8 for its count, with a 16-bit file header. What percentage of original bits is saved?

Test 3 · 20 marks

T3Q01 · NAT · 2 marks

Customer has unique IDs 1,2,3. Purchase references are 2,2,NULL, with valid non-null references. Count rows of Customer LEFT JOIN Purchase on matching IDs.

T3Q02 · MSQ · 2 marks

R(A,B) and S(B,C) are set relations. Select all true statements. A: selection C=5 on R natural-join S can be pushed into S. B: projecting away B from R before joining always preserves the query. C: set projection removes duplicate projected tuples. D: a natural join only matches the first shared attribute if several are shared.

T3Q03 · MSQ · 2 marks

Select all true statements in standard SQL. A: COUNT(v) ignores NULL. B: COUNT(*) ignores NULL-containing rows. C: EXISTS(SELECT NULL) is TRUE. D: NULL=NULL is TRUE.

T3Q04 · NAT · 2 marks

F={A→B,B→C,A→C} on ABC. What is the number of dependencies in a singleton-right-side minimal cover?

T3Q05 · MCQ · 2 marks

R(ABCD), with AB→CD,C→B,C→D, is split into BCD and AC. Which statement is correct? A: lossless and preserving. B: lossy and preserving. C: lossless but not preserving. D: lossy and not preserving.

T3Q06 · MCQ · 2 marks

A has 4 entities and B has 5. A→B is many-to-one and both sides are total. Which is correct? A: feasible with 4 pairs. B: feasible with 5 pairs. C: infeasible. D: feasible with 20 pairs.

T3Q07 · NAT · 2 marks

A B+ tree consists of one leaf [2,4] of capacity 2. Insert 6, splitting into 2 entries left and 1 right and using the minimum of the right leaf as separator. What key appears in the new root?

T3Q08 · NAT · 2 marks

Three non-null dimensions have only two observed detail combinations: (A,X,1),(B,X,1). Count total rows across all CUBE grouping sets.

T3Q09 · NAT · 2 marks

Population strata have sizes 75 and 25. Their sample means are 8 and 20. Compute the stratified estimate of the population mean.

T3Q10 · MCQ · 2 marks

Which measurement has a meaningful zero and meaningful ratios? A: postal code. B: ordered satisfaction category. C: Celsius temperature. D: elapsed time in seconds.

Test 4 · 20 marks

T4Q01 · NAT · 2 marks

A seven-attribute relation has exactly one candidate key containing three attributes. Count its superkeys.

T4Q02 · NAT · 2 marks

X has P values 1,2,3, one row each. Y(P,S)={(1,4),(1,5),(2,0),(3,6)}. The output contains each x with some same-P Y row satisfying S>0. How many output rows, not witness pairs, occur?

T4Q03 · NAT · 2 marks

Employee rows (ID,Team) are (1,A),(2,A),(3,A),(4,A),(5,B),(6,B). A self join on equal Team groups by the left ID and counts right rows. What is the largest resulting count?

T4Q04 · MSQ · 2 marks

Atomic R(ABCD) has AB→C,C→B,A→D. Select all true statements. A: it is 1NF but not 2NF. B: it is BCNF. C: AC is a candidate key. D: C is a candidate key.

T4Q05 · MCQ · 2 marks

R(ABCD) has A→B,B→C,C→D and is split into AB,BC,CD. Which is correct? A: lossless and preserving. B: lossy and preserving. C: lossless but not preserving. D: lossy and not preserving.

T4Q06 · NAT · 2 marks

Two strong entities have a one-to-one relationship with total participation on both sides. Each has a single-attribute key and single-valued descriptive attributes; one also has one multivalued attribute. Merging their single-valued rows is allowed. What is the minimum number of atomic 3NF relations?

T4Q07 · NAT · 2 marks

An uncached B+ point lookup traverses 4 levels. All requested values are present in the leaf index entry. How many page reads are needed?

T4Q08 · NAT · 2 marks

One group has count 3 and sum 60; another has count 7 and sum 70. Find the mean of all observations.

T4Q09 · NAT · 2 marks

Map x=25 from original range [10,30] to [0,100] by min-max normalization.

T4Q10 · MSQ · 2 marks

Select all true statements. A: lossless compression shortens every input. B: encoded size must include its header. C: replacing observations by bin means always allows exact reconstruction. D: run-length encoding can expand a file.